KLAi

Solving linear systems: from Ax = b to the best-fitting line

How does a computer find the best straight line through data in one step, and why does a model with more features than rows break?

Mathematics for AI and Machine Learning | Session 4 of 18 | Linear algebra
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Write a system of equations as Ax = b and solve it by elimination and with the inverse

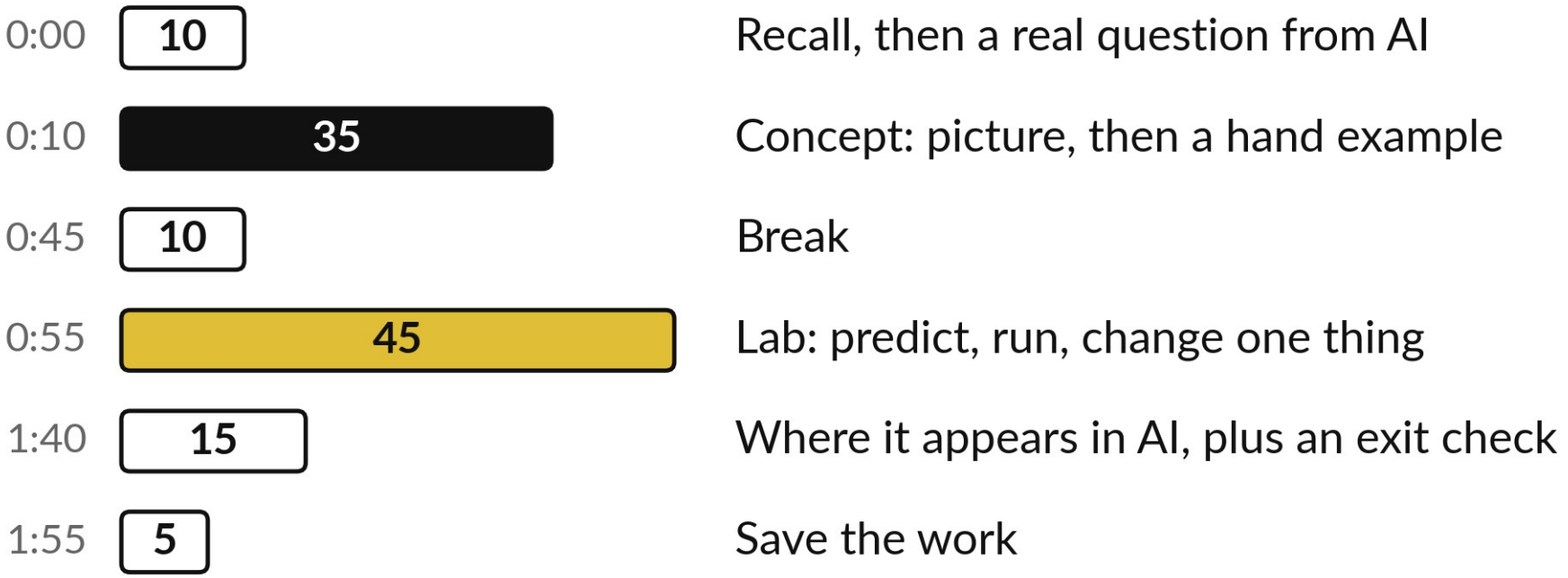


Say what the determinant and the rank tell you, and when a system has no solution or many



Fit a line $\hat{y} = b + w\,x$ by least squares with the normal equation, in Python and in a spreadsheet

Your two hours: equations, then a fitted line



Ideas first, with pictures. Then the lab in Python or in a spreadsheet. Then where it shows up in AI.

Three questions from last time

1. A is a 3 by 2 matrix and B is a 2 by 4 matrix. Can you multiply A times B, and what is the shape of the result?

2. What does the matrix with rows (2, 0) and (0, 3) do to the vector (1, 1)?

3. What does the identity matrix I do to any vector x?

Answer aloud first. Then we check together.

Answers to the three questions from Session 3

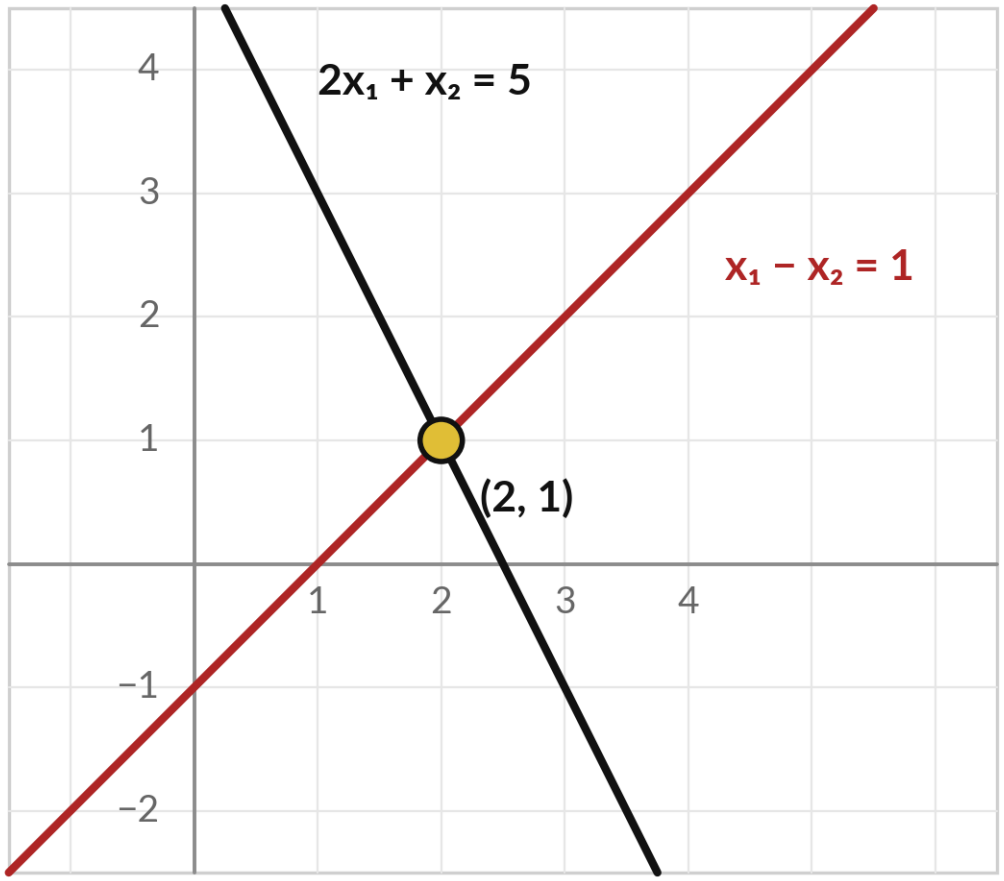
Question	Short answer
A is a 3 by 2 matrix and B is a 2 by 4 matrix. Can you multiply A times B, and what is the shape of the result?	Yes. The inner numbers match (2 and 2). The result is 3 by 4.
What does the matrix with rows (2, 0) and (0, 3) do to the vector (1, 1)?	It gives (2, 3). It stretches the first direction by 2 and the second by 3.
What does the identity matrix I do to any vector x?	Nothing. I times x equals x. It is the matrix version of the number 1.

If one surprised you, look back at Session 3 before the lab. The identity matrix returns today as part of the inverse.

Words we use today

	Word	What it means
	**Linear system**	Several equations in the same unknowns, written together as Ax = b.
	**Inverse**	The matrix A^{-1} that undoes A: A^{-1} times A is the identity matrix.
	**Determinant**	One number from a square matrix. Its size is the factor by which A scales areas.
	**Rank**	How many independent directions the columns of a matrix give: no chosen column can be built from the others.
	**Least squares**	Choosing the line whose squared errors add up to the smallest total.

A system of equations is a set of lines, and the solution is where they cross



Two equations, two unknowns, x_1 (across) and x_2 (up). Each equation draws one line.

The gold dot lies on both lines, so it satisfies both equations.

It is the pair $x_1 = 2$, $x_2 = 1$.

Check: $2 \times 2 + 1 = 5$ and $2 - 1 = 1$.

The same system in matrix form is Ax = b

Matrix form

$$A\,\mathbf{x} = \mathbf{b}$$

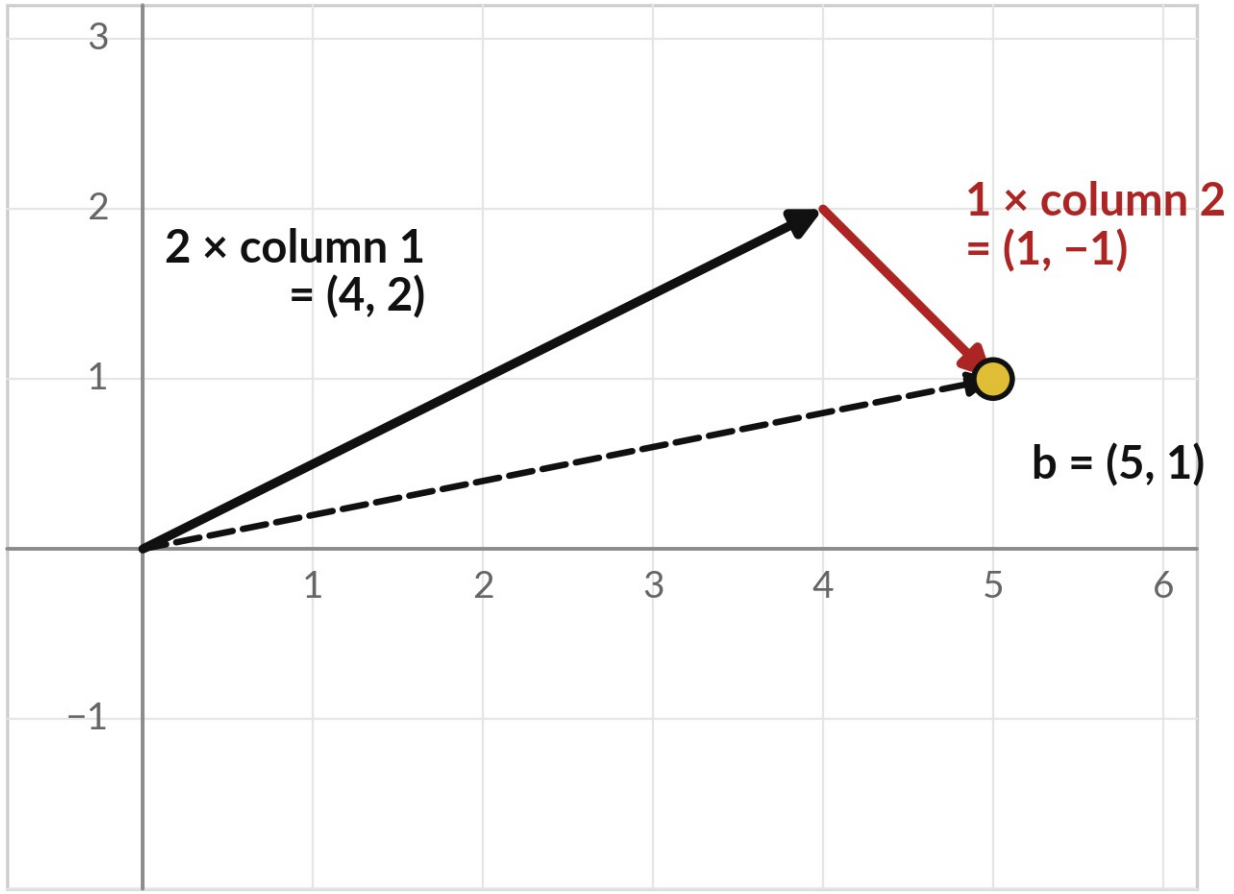
Our system

$$2x_1 + x_2 = 5 \qquad x_1 - x_2 = 1$$

Symbol	What it means	In our data
A	The matrix of coefficients. Row 1 is (2, 1) and row 2 is (1, −1).	2 by 2 matrix
x	The vector of unknowns, a column (Session 3). Its entries are x_1 and x_2.	(x_1, x_2)
b	The vector of right-hand sides. Here b is the right-hand side; from the line fit on, b is the intercept.	(5, 1)
Ax	Matrix times vector from Session 3: each row of A takes a dot product with x.	$(2x_1 + x_2, x_1 - x_2)$

Ax = b says: apply A to the unknown x and you get b. Solving means running A backwards.

The column picture: b is a mix of the columns of A



The columns of A are the arrows (2, 1) and (1, -1). How much of each adds up to b?

2 of column 1 reaches (4, 2). Then 1 of column 2 moves on to (5, 1).

That is b. So the solution $(x_1, x_2) = (2, 1)$ lists how much of each column to use.

Elimination solves the system by making zeros, then reading the answer from the bottom

Start: the augmented matrix

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array}\right]$$

The dashed line separates A from b. Each row is one equation

Row 2 minus half of row 1

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ \mathbf{0} & \mathbf{-1.5} & \mathbf{-1.5} \end{array}\right]$$

a row operation: the 1 under the 2 becomes 0

Divide row 2 by −1.5

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 0 & 1 & 1 \end{array}\right]$$

row 2 says $x_2 = 1$. Then row 1 says $2x_1 + 1 = 5$, so $x_1 = 2$

Making zeros, then back substitution (bottom row first): together, Gaussian elimination.

The inverse undoes A, so x = A^{-1}b

Inverse and solution

$$A^{-1}A = I \qquad \mathbf{x} = A^{-1}\mathbf{b}$$

For a 2 by 2 matrix

$$A^{-1} = \frac{1}{\det A}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Symbol	What it means	In our data
A^{-1}	The inverse of A: the matrix that undoes A.	exists only if det A is not 0
I	The identity matrix, which leaves every vector unchanged.	rows (1, 0) and (0, 1)
a_{11} to a_{22}	The entries of A: row 1 is (a_{11}, a_{12}) and row 2 is (a_{21}, a_{22}). The inverse swaps a_{11} and a_{22} and puts a minus sign on the other two.	2, 1, 1, −1
det A	The determinant of A: $a_{11}a_{22} - a_{12}a_{21}$.	2 × (−1) − 1 × 1 = −3

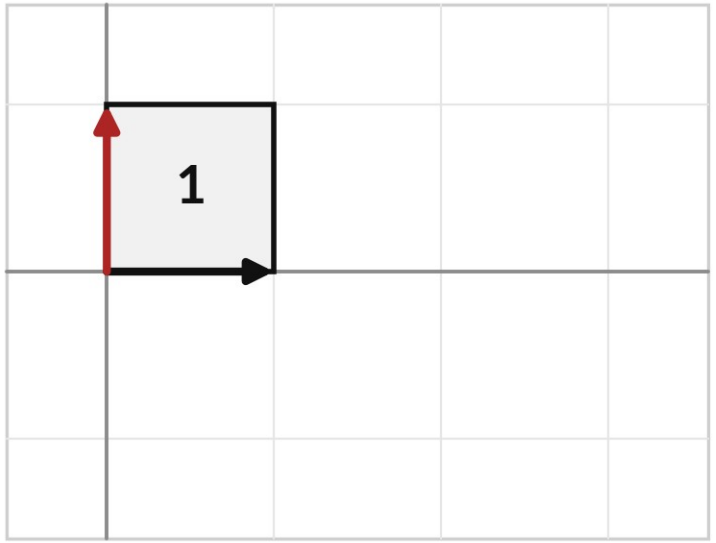
If x = A^{-1}b, then Ax = AA^{-1}b = Ib = b. The answer undoes the system.

Worked example: the inverse of our A, and the solution from it

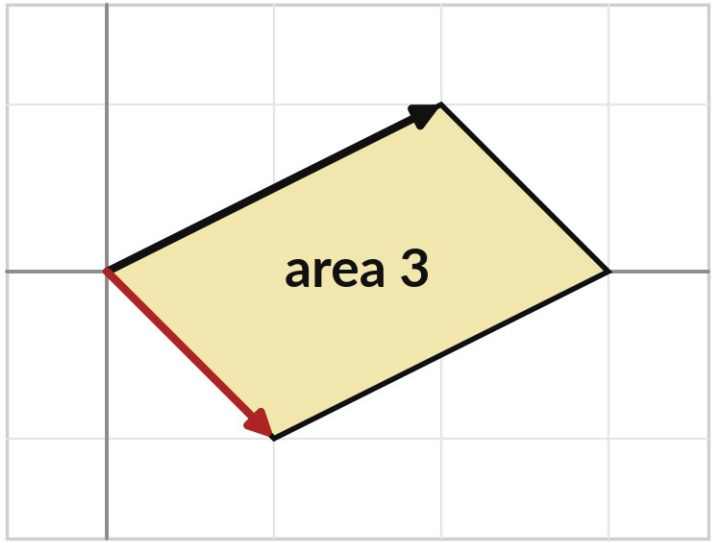
A has rows (2, 1) and (1, −1). We find the inverse, then multiply it by b = (5, 1).

Determinant	det A = 2 × (−1) − 1 × 1 = −3. It is not 0, so the inverse exists.
Inverse	Swap 2 and −1, negate the other two: rows (−1, −1) and (−1, 2). Divide by −3: rows (1/3, 1/3) and (1/3, −2/3).
Solution	$x = A^{-1}b$: first entry (5 + 1) / 3 = 2. Second entry (5 − 2) / 3 = 1.
Check	A times (2, 1) gives (2 × 2 + 1, 2 − 1) = (5, 1). It equals b.

The determinant is the factor by which A scales areas



Unit square: area 1



A turns it into this: area 3

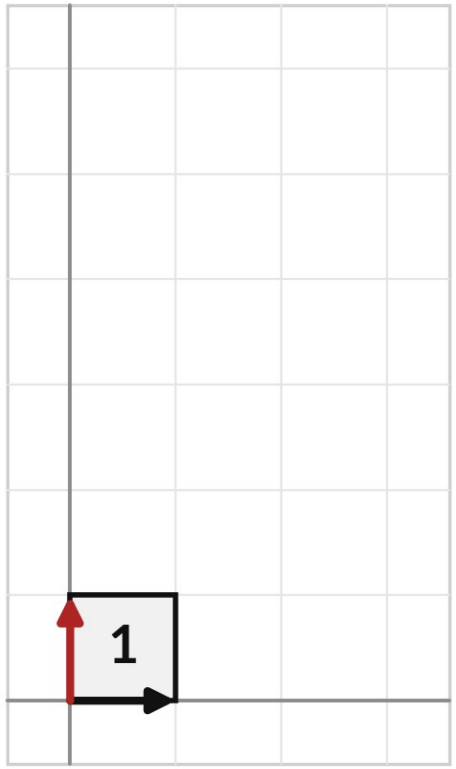
The two red and black arrows are the columns of A. A sends the unit square onto the gold parallelogram.

Its area is 3, so A scales areas by 3.

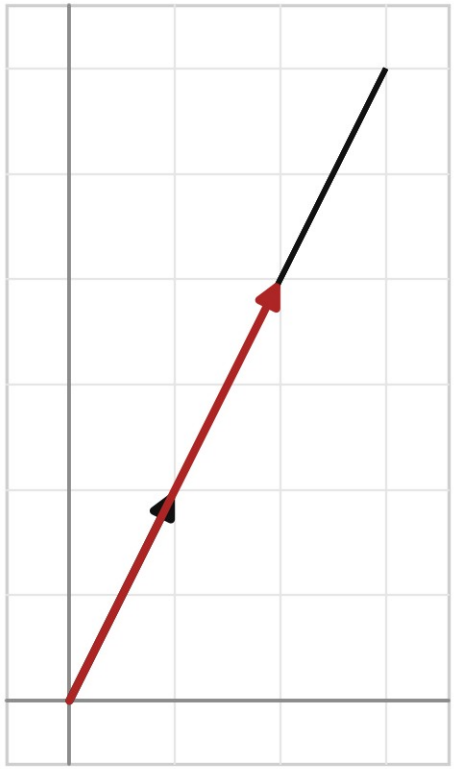
det A = −3. The size is the area factor.

The minus sign means the picture is also flipped, like a mirror.

When the determinant is 0, A squashes the plane flat and has no inverse



Unit square: area 1



A turns it into a line: area 0

This matrix has columns (1, 2) and (2, 4). Both arrows point the same way.

The square collapses onto one line, so the area is 0.

$\det = 1 \times 4 - 2 \times 2 = 0.$

Many inputs land on the same output, so nothing can undo it: no inverse. Such a matrix is called singular.

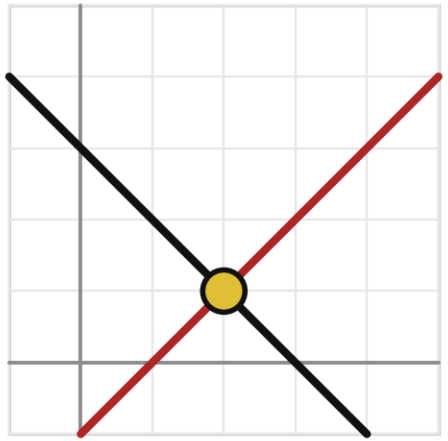
Rank counts how many independent directions the columns give

Matrix	Determinant	Rank	What the columns can reach
Our A: columns (2, 1) and (1, -1)	−3	2	The whole plane
Singular: columns (1, 2) and (2, 4)	0	1	Only one line
Zero matrix: all entries 0	0	0	Only the origin

Rank = the largest number of columns you can choose so that none of them can be built from the others. A 2 by 2 matrix has an inverse exactly when its rank is 2.

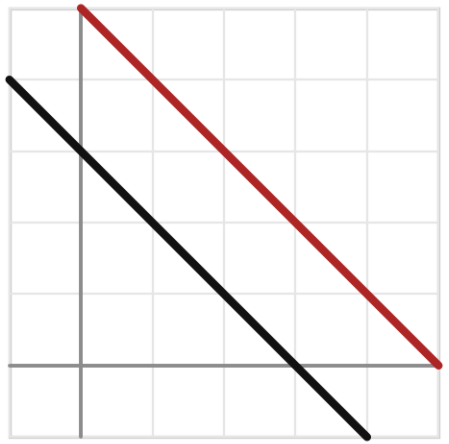
A system has one solution, none, or many, and rank and b decide which

One solution



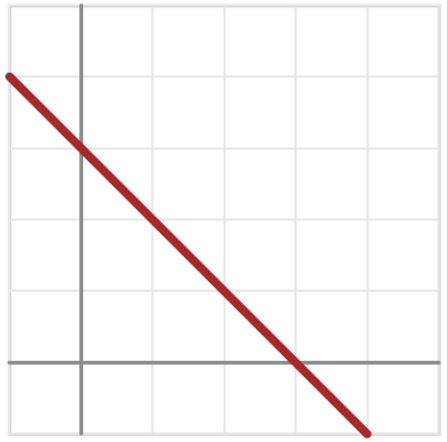
$x_1 + x_2 = 3$
$x_1 - x_2 = 1$

No solution



$x_1 + x_2 = 3$
$x_1 + x_2 = 5$

Many solutions



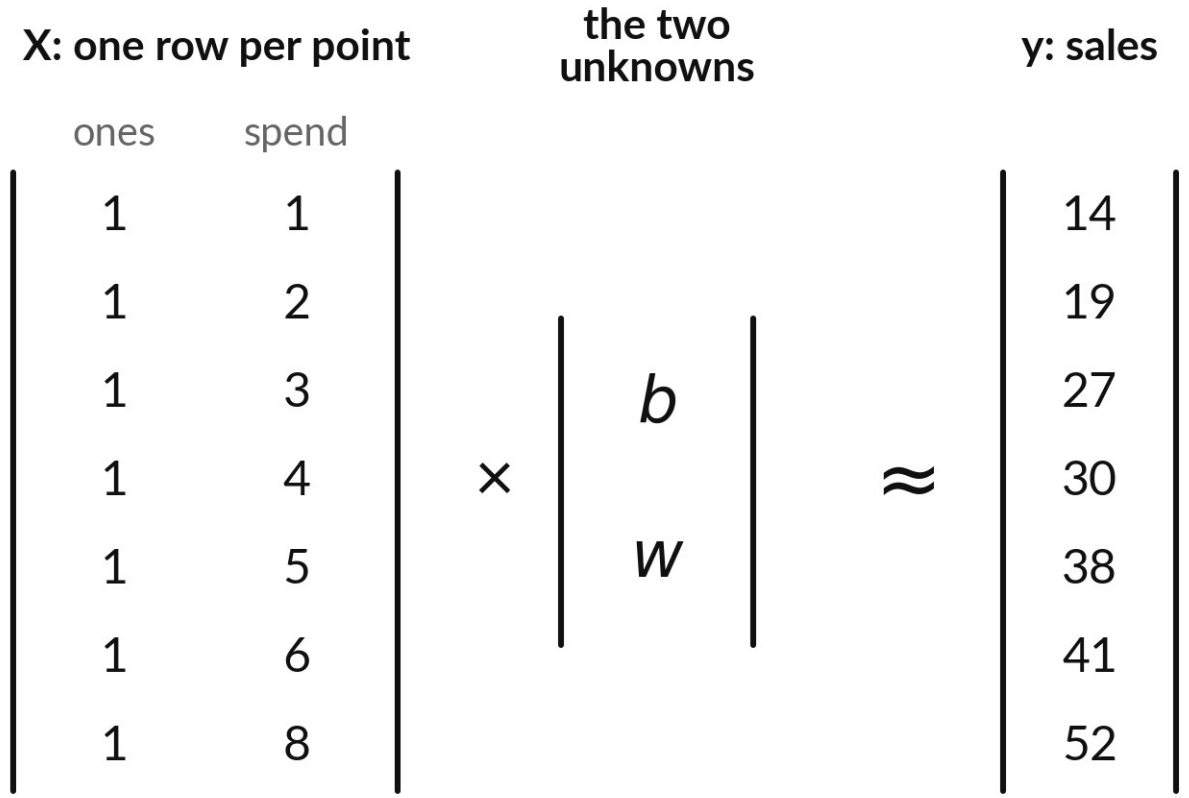
$x_1 + x_2 = 3$
$2x_1 + 2x_2 = 6$

Lines that cross once give one solution. Parallel lines never meet. The same line twice meets everywhere.

Full rank: exactly one solution.

Rank below full: no solution or infinitely many, depending on b.

Fitting a line to 7 data points is 7 equations with only 2 unknowns



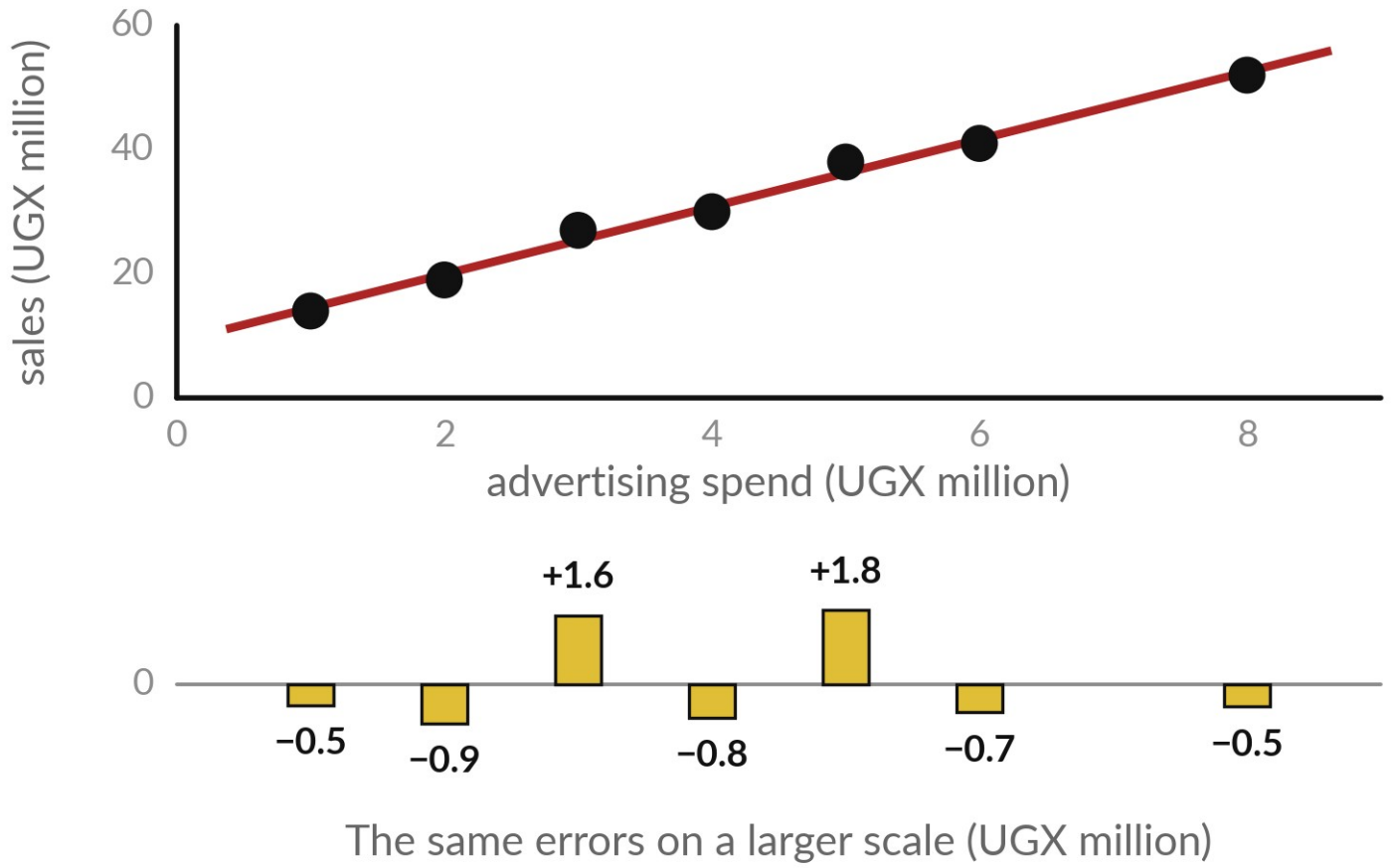
Each row says b + w × spend = sales, for one month. Seven rows, two unknowns: the intercept b and the slope w.

From here on, b and w are the unknowns and X holds the data.

The column of ones in X carries b.

The points are not on one line, so no exact solution exists.

Least squares picks the line with the smallest sum of squared errors



Each dot is one month. The gold bars show its error, also called the residual: actual sales y minus the line's prediction ŷ.

Line: $\hat{y} = 9.06 + 5.43 \times \text{spend}$.

Least squares makes the sum of squared errors smallest: 8.28.

RMSE (root mean squared error) = $\sqrt{(8.28 \div 7)} = 1.09$.

Errors at right angles to the columns of X give the normal equation

Errors at right angles $$X^T(\mathbf{y} - X\mathbf{w}) = \mathbf{0}$$

Normal equation $$(X^T X)\,\mathbf{w} = X^T\mathbf{y}$$

Symbol	What it means	In our data
X	The data: a column of ones, then spend. Tall: more rows than unknowns.	7 rows, 2 columns
y, w	y: the sales. w (bold): the unknowns in a column, b on top, w below.	7 sales; (b, w)
y - Xw	The errors: actual minus predicted.	7 errors
X^T(...) = 0	Each column of X has dot product 0 with the errors: right angles (Session 2).	2 equations
X^TX	Square, one row per unknown. If it has an inverse, $w = (X^TX)^{-1}X^Ty$.	2 by 2

Multiply out line 1: $X^Ty - X^TXw = 0$, so $X^TXw = X^Ty$.

Worked by hand: the normal equation for 7 points

Spend 1, 2, 3, 4, 5, 6, 8 and sales 14, 19, 27, 30, 38, 41, 52 (UGX million, illustrative).

X^TX and X^Ty	X^TX: rows (7, 29) and (29, 155), count, sums of spend and spend squared. X^Ty = (221, 1105).
Inverse	det = 7 × 155 − 29 × 29 = 244. Swap and negate: (1/244) × rows (155, −29) and (−29, 7).
Multiply	Times (221, 1105): b = (155 × 221 − 29 × 1105) / 244 = 9.06. w = (−29 × 221 + 7 × 1105) / 244 = 5.43.
Meaning	Each extra million of spend goes with 5.43 million more sales. At spend 7: 47.1.

Two ways to solve: the inverse, or a solver

Inverse

$x = A^{-1}b$ is the idea to understand.

Computing the whole inverse takes more work.

For large matrices it is usually less accurate.

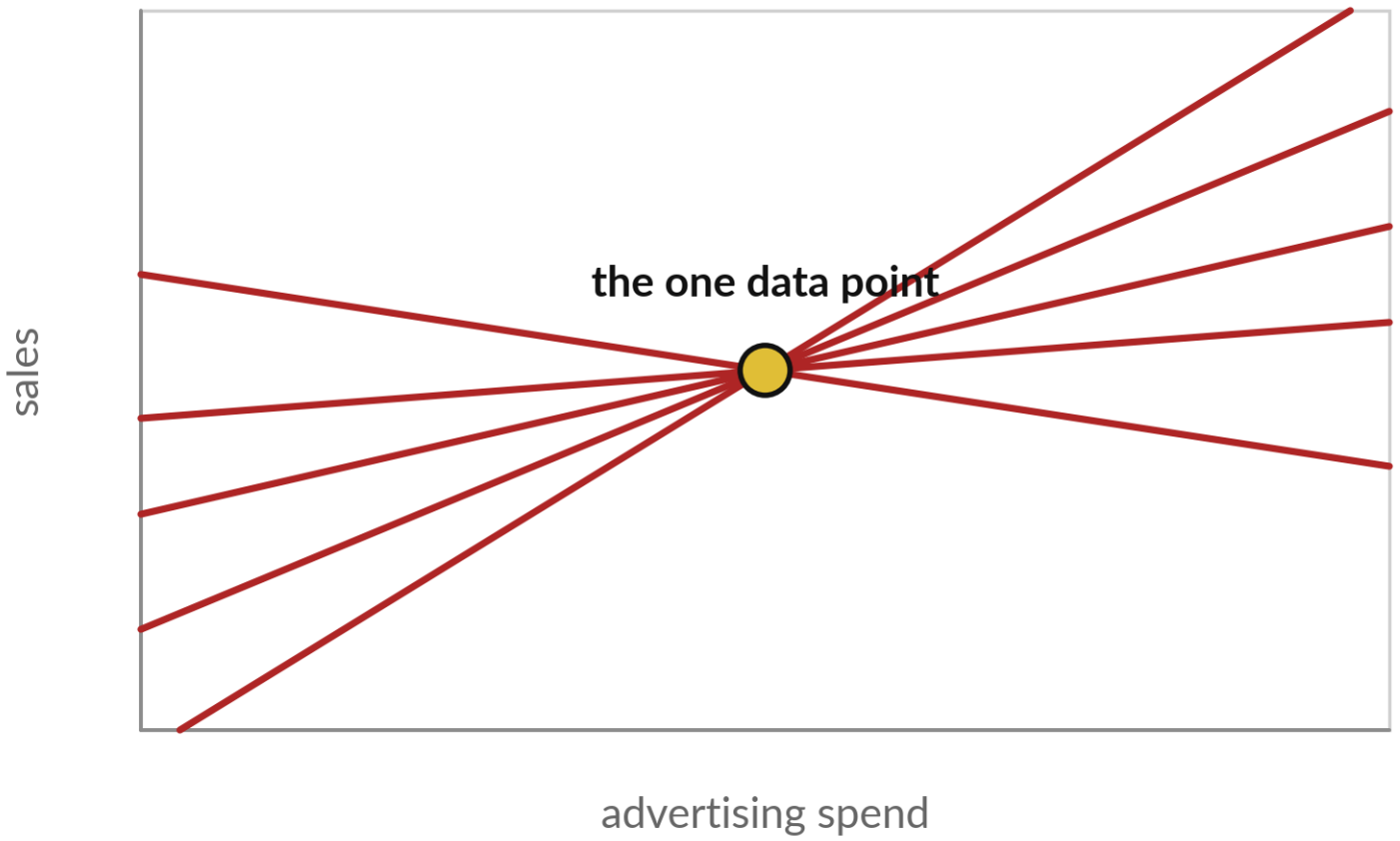
Solvers

solve(A, b) factorises A, a form of elimination.

lstsq(X, y) fits by least squares, using the SVD (Session 6).

Learn the inverse to understand it. Use a solver to compute.

Too many unknowns and too few rows: infinitely many lines fit perfectly



One data point and two unknowns, the intercept b and the slope w. Every red line through the point fits it exactly.

With spend 4: X^TX has rows (1, 4) and (4, 16), and determinant $1 \times 16 - 4 \times 4 = 0$.

The rank of X is at most its number of rows, here 1, so the rank of X^TX is at most 1. No inverse, so the normal equation has many answers.

Four mistakes that break a solve, and what to do instead

Mistake	Why it fails	What to do
Dividing by a matrix	There is no matrix division.	Multiply by the inverse, on the left: $A^{-1}b$.
Inverting when the determinant is 0	A singular matrix has no inverse.	Check the determinant or the rank first.
Leaving out the column of ones	The line is forced through the origin.	Add a column of 1s to X for the intercept b.
Swapping the order of a product	AB and BA usually differ.	Keep A^{-1} on the left of b.

Six NumPy names cover today: solve, det, inv, matrix_rank, lstsq and the transpose

```
x = np.linalg.solve(A, b)
d = np.linalg.det(A)
A_inv = np.linalg.inv(A)
r = np.linalg.matrix_rank(S)
X = np.column_stack([np.ones(7), spend])
bw = np.linalg.solve(X.T @ X, X.T @ sales)
bw2 = np.linalg.lstsq(X, sales, rcond=None)[0]
```

In plain words

solve, det, inv and matrix_rank do what their names say: solve Ax = b, the determinant, the inverse, the rank.

X = ... puts a column of ones next to spend.

bw = ... is the normal equation. X.T is the transpose, @ is matrix multiplication. bw holds b, then w.

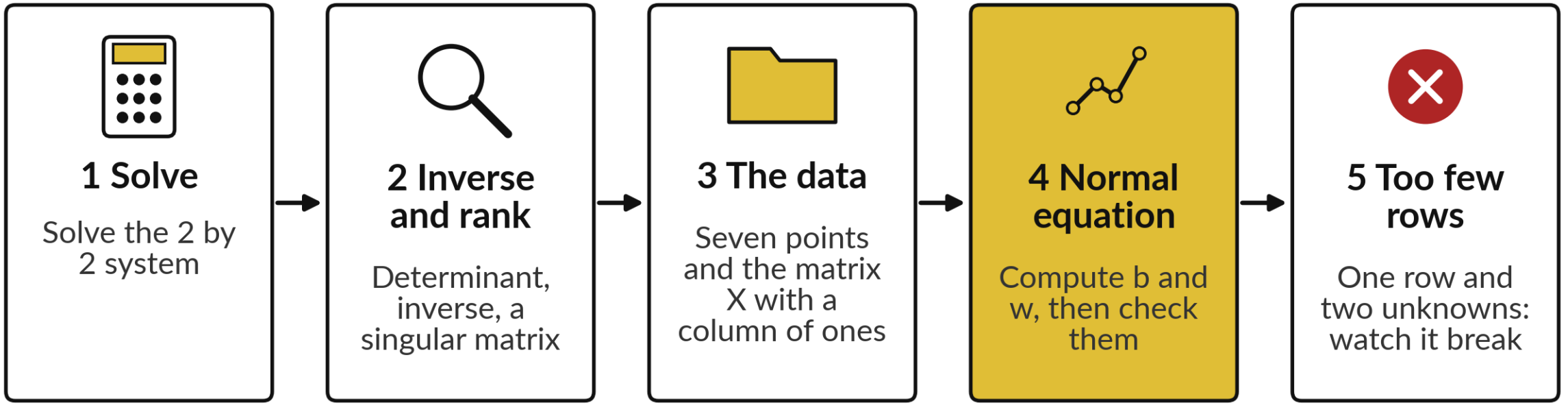
bw2 = ... is least squares in one call. rcond=None silences a warning, and the final 0 picks the weights.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

Five lab steps take you from a 2 by 2 solve to a fitted line



Python uses np.linalg. The spreadsheet uses MMULT, MINVERSE, MDETERM and TRANSPOSE as array formulas. Gold cells are yours.

Your numbers should match: 2 and 1, then b = 9.06 and w = 5.43

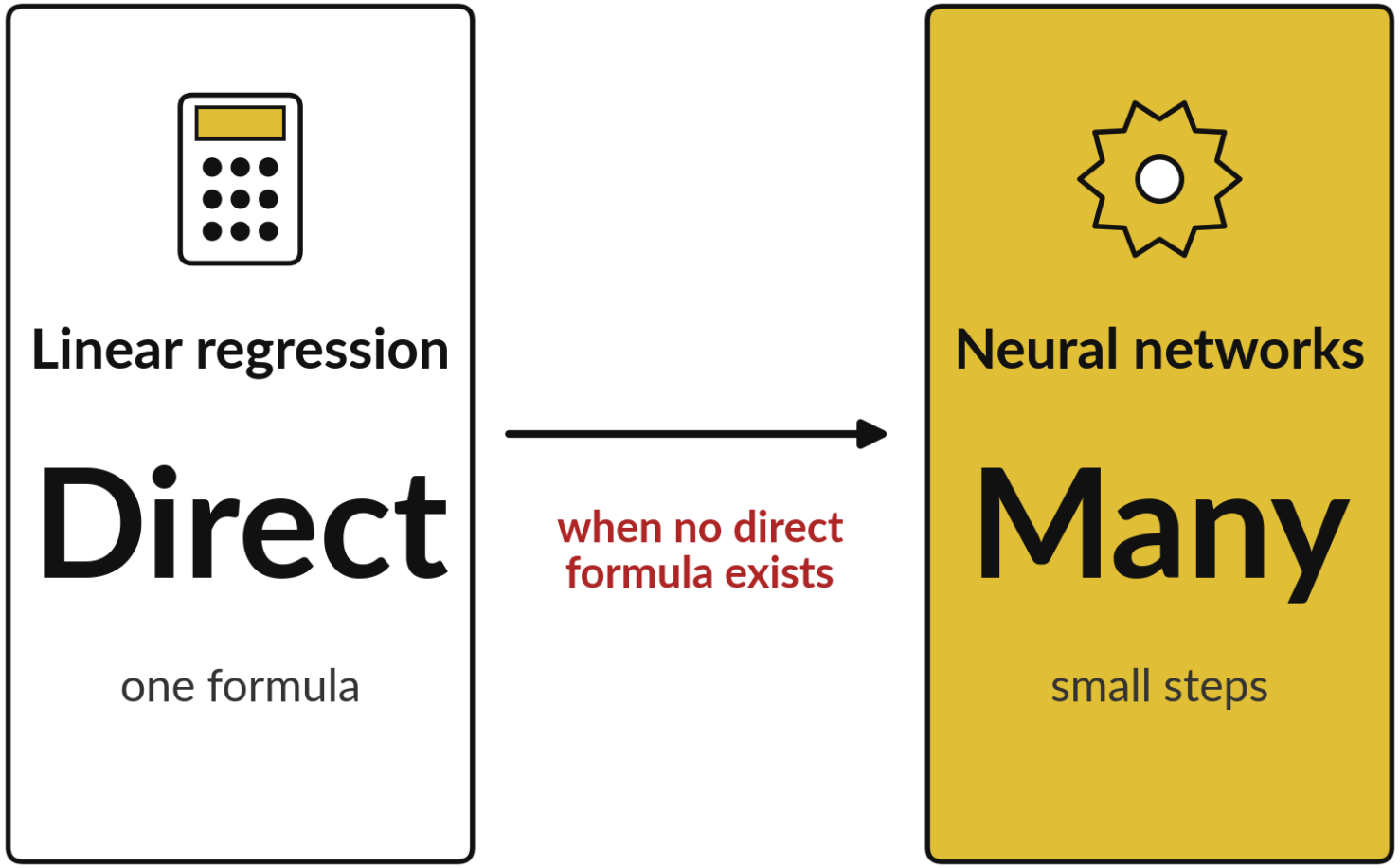
	What you compute	You should see
🧮	**Solution x_1 and x_2 of the 2 by 2 system**	2 and 1
🔑	**Determinant of A**	−3
🔑	**Top-left entry of the inverse of A**	0.33 (that is 1/3)
🚩	**Rank of the singular matrix (rows (1, 2) and (2, 4))**	1
▦	**X^TX for the seven points**	rows (7, 29) and (29, 155)
📈	**Intercept b and slope w**	9.06 and 5.43
📈	**Total squared error, then RMSE**	8.28, then 1.09 (UGX million)
❌	**Determinant of X^TX with one row**	0

If a number differs, stop and find the step that went wrong. Do not type a number to make a check green.

Singular matrices, shapes and array formulas cause most lab errors

Error or symptom	What it says	What to do
LinAlgError: Singular matrix	The matrix has determinant 0, so no inverse.	Check the rank. Use lstsq instead.
ValueError: shapes not aligned	The inner sizes of a product do not match.	Print .shape of both and compare.
#SPILL! or #N/A from an array formula	Excel 365: cells below are not empty. Older Excel: the selected range has the wrong size.	Clear the cells, or select the exact range. Older Excel: Ctrl+Shift+Enter. Excel 365 and Sheets: Enter.
Intercept b comes out 0	The column of ones is missing from X.	Add the column of ones.

Where this appears in AI: solving in one step, or improving in small steps



Linear regression has a closed-form answer. Models of this kind learn in many small updates.

Published work trains them with Adam, in small steps.

Transformer layers (Vaswani et al., 2017) are mostly matrix multiplications.

Too many unknowns: add a penalty (ridge, Session 12).

Where this appears in AI: ask an assistant for the line, then check it

8 minutes

Hands-on, with any AI assistant

Ask an assistant to fit the line, then check it against yours.

Type the seven spend and sales values from the worked example. Never paste confidential data.

Ask for a least squares line with the normal equation steps.

Compare its intercept and slope with 9.06 and 5.43.

Compare its determinant of X^TX with 244.

You keep

A note of where the assistant agrees with your numbers, and where it differs.

Exit check: the determinant, the column of ones, and too few rows

Question	Short answer
What does a determinant of 0 tell you about Ax = b?	A has no inverse. There is no solution or infinitely many, depending on b.
Why do we need a column of ones in X?	It carries the intercept b. Without it the line passes through the origin.
Why does one data point and two unknowns break the normal equation?	X^TX has determinant 0 and rank 1, so many lines fit equally well.

Write your own answer first, then compare. Your lab checks are part of the exit check.

Save the lab and keep the fitted line

What you keep from today

Keep your saved notebook or workbook, and the fitted line for the advertising data: intercept b = 9.06 and slope w = 5.43.

Our question today

How does a computer find the best straight line through data in one step, and why does a model with more features than rows break?

Next session

Session 5: eigenvalues and eigenvectors, the directions a matrix only stretches.

Take away

Saved lab
Notebook or workbook, saved

Fitted line
$\hat{y} = 9.06 + 5.43 \times spend$

Next session
Eigenvalues and eigenvectors

Bring
One question about today